2012 A level H2 Paper 1 Suggested Solutions

1	Solutions
	L et x, y and z be the cost of a ticket for each of the groups under 16 years, between 16 and 65 years, and over 65 years respectively. Always define constants and variables clearly. $9x+6y+4z=162.03$ ...(1) $7x+5y+3z=128.36$ ...(2) $10x+4y+5z=158.50$ ...(3) Using GC, $x=7.65$, $y=9.85$, $z=8.52$ The cost of a ticket for each of the groups under 16 years, between 16 and 65 years, and over 65 years are \$7.65, \$9.85 and \$8.52 respectively.

2	Solutions
(i)	$\int \frac{x^3}{1+x^4}\,dx = \frac{1}{4}\int \frac{4x^3}{1+x^4}\,dx$ $= \frac{1}{4}\ln\left(1+x^4\right)+C$, where C is a constant **Formula (not in MF26):** $\int \frac{f'(x)}{f(x)}\,dx = \ln\lvert f(x)\rvert + C$ **Note:** $\left(1+x^4\right)>0$ for all $x\in\mathbb{R}$, $\therefore$ modulus not required
(ii)	$u=x^2$ $\frac{du}{dx}=2x$ $\int \frac{x}{1+x^4}\,dx = \frac{1}{2}\int \frac{1}{1+u^2}\,du$ $=\frac{1}{2}\tan^{-1}(u)+C$ $=\frac{1}{2}\tan^{-1}\left(x^2\right)+C$ Final answer in original variable x **Formula (in MF26):** $\int \frac{1}{x^2+a^2}\,dx = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right)+C$
(iii)	$\int_0^1 \left(\frac{x}{1+x^4}\right)^2 dx = 0.186$ (3 d.p.) When the question asks for numerical answer for integration, always use GC.

Qn 3 Out of syllabus (9758)

4	Solutions
(i)	By sine rule, $\frac{AC}{\sin\left(\frac{3}{4}\pi\right)} = \frac{1}{\sin\left(\pi-\frac{3}{4}\pi-\theta\right)}$ $AC = \frac{\sin\left(\frac{3}{4}\pi\right)}{\sin\left(\frac{1}{4}\pi-\theta\right)}$ We use **sine rule** as there are two angles and the length of one side provided, whereas for cosine rule, we would need the lengths of two sides and one angle.

$$AC = \frac{\frac{1}{\sqrt{2}}}{\sin\left(\frac{1}{4}\pi\right)\cos\theta - \cos\left(\frac{1}{4}\pi\right)\sin\theta}$$

Apply Addition Formula (MF26) – Pg 3, 1st line.

$$AC = \frac{\frac{1}{\sqrt{2}}}{\frac{1}{\sqrt{2}}\cos\theta - \frac{1}{\sqrt{2}}\sin\theta}$$

$$AC = \frac{1}{\cos\theta - \sin\theta} \text{ (shown)}$$

(ii) For small θ,

$$AC \approx \frac{1}{1-\frac{\theta^2}{2}-\theta}$$

Apply Small Angle Approximation:

$\sin\theta \approx \theta$, $\cos\theta \approx 1-\frac{\theta^2}{2}$, $\tan\theta \approx \theta$.

$$AC \approx \left[1-\left(\theta+\frac{\theta^2}{2}\right)\right]^{-1}$$

$$= 1+\left(\theta+\frac{\theta^2}{2}\right)+\left(\theta+\frac{\theta^2}{2}\right)^2+\cdots$$

$$= 1+\left(\theta+\frac{\theta^2}{2}\right)+\left(\theta^2+2\theta\left(\frac{\theta^2}{2}\right)+\cdots\right)+\cdots$$

Apply Binomial series formula (under Maclaurin's expansion), where $x = -\left(\theta+\frac{\theta^2}{2}\right)$.

$$= 1+\theta+\frac{3}{2}\theta^2+\cdots$$

$$\therefore a = 1,\ b = \frac{3}{2}$$

5	Solutions

(i)

$$\overrightarrow{OA} = \mathbf{a} = \begin{pmatrix}1\\-1\\1\end{pmatrix},\quad \overrightarrow{OB} = \mathbf{b} = \begin{pmatrix}1\\2\\0\end{pmatrix},\quad \overrightarrow{OC} = \lambda\mathbf{a}+\boldsymbol{\mu}b = \begin{pmatrix}\lambda+\mu\\-\lambda+2\mu\\\lambda\end{pmatrix}$$

Area of triangle OAC = $\sqrt{126}$

$$\frac{1}{2}\left|\overrightarrow{OA}\times\overrightarrow{OC}\right| = \sqrt{126}$$

$$\left|\begin{pmatrix}1\\-1\\1\end{pmatrix}\times\begin{pmatrix}\lambda+\mu\\-\lambda+2\mu\\\lambda\end{pmatrix}\right| = 2\sqrt{126}$$

$$\left|\begin{pmatrix}-2\mu\\\mu\\3\mu\end{pmatrix}\right| = 2\sqrt{126}$$

$$\sqrt{4\mu^2+\mu^2+9\mu^2} = 2\sqrt{126}$$

$$14\mu^2 = 504$$

Alternatively: $\mathbf{a}\times\lambda\mathbf{a} = \mathbf{0}$ for any vector $\mathbf{a}$, hence $\mathbf{a}\times(\lambda\mathbf{a}+\mu\mathbf{b}) = \mathbf{a}\times\mu\mathbf{b} = \mu(\mathbf{a}\times\mathbf{b})$.

$$\therefore \left|\overrightarrow{OA}\times\overrightarrow{OC}\right| = |\mathbf{a}\times\mathbf{c}| = |\mu(\mathbf{a}\times\mathbf{b})|$$

$$= \mu\left|\begin{pmatrix}1\\-1\\1\end{pmatrix}\times\begin{pmatrix}1\\2\\0\end{pmatrix}\right| = \mu\left|\begin{pmatrix}-2\\1\\3\end{pmatrix}\right|$$

	$\mu^2 = 36$ Since $\mu > 0$, we have $\mu = 6$.
(ii)	Given $\mu = 4$, $OC = 5\sqrt{3}$ $\left\| \begin{pmatrix} \lambda + 4 \\ -\lambda + 2(4) \\ \lambda \end{pmatrix} \right\| = 5\sqrt{3}$ $\sqrt{(\lambda+4)^2 + (8-\lambda)^2 + \lambda^2} = 5\sqrt{3}$ $3\lambda^2 - 8\lambda + 80 = 75$ $3\lambda^2 - 8\lambda + 5 = 0$ $(3\lambda - 5)(\lambda - 1) = 0$ $\lambda = 1$ or $\lambda = \frac{5}{3}$ $\therefore C(5,7,1)$ or $C\left(\frac{17}{3}, \frac{19}{3}, \frac{5}{3}\right)$ Give coordinates as required by question. **Note:** Question says "possible coordinates", indicating there is more than 1 answer.

6	**Solutions**
(i)	$z^3 = (1 + \mathrm{i}c)^3$ $= 1^3 + \binom{3}{1} 1^2 (\mathrm{i}c) + \binom{3}{2} 1 (\mathrm{i}c)^2 + (\mathrm{i}c)^3$ ← Apply binomial expansion formula (see MF26). $= 1 + 3\mathrm{i}c + 3(\mathrm{i}c)^2 + (\mathrm{i}c)^3$ $= 1 + 3\mathrm{i}c - 3c^2 - \mathrm{i}c^3$ $= 1 - 3c^2 + \mathrm{i}(3c - c^3)$ ← Put in the form $x + \mathrm{i}y$ as required by question.
(ii)	z^3 is real $\Rightarrow \mathrm{Im}(z^3) = 0 \Rightarrow 3c - c^3 = 0$ $c(3 - c^2) = 0$ $c^2 = 3$ since c is non-zero $c = \pm\sqrt{3}$ $\therefore z = 1 + \mathrm{i}\sqrt{3}$ or $1 - \mathrm{i}\sqrt{3}$
(iii)	Let $z = 1 - \mathrm{i}\sqrt{3}$ Since $\left\|z^n\right\| = \left\|z\right\|^n = \left\|1 - \mathrm{i}\sqrt{3}\right\|^n = 2^n$, $\left\|z^n\right\| > 1000 \Rightarrow 2^n > 1000$ Since $2^9 = 512 < 1000$ $2^{10} = 1024 > 1000$ Show sign change least n is 10. Not allowed to take 'ln' for both sides as a calculator **CANNOT** be used in answering this question. **Note:** Can use G.C. to double check your answer, so just need to show proper working in solutions.

$$\left|z^{10}\right| = 1024$$

$$\arg\left(z^{10}\right) = 10\arg(z) = 10\left(-\frac{\pi}{3}\right) \equiv \frac{2\pi}{3}$$

Convert to principal argument range of $(-\pi, \pi]$

7	**Solutions**

(i)

Let $y = \dfrac{x+k}{x-1}$

$xy - y = x + k$

$x(y-1) = y + k$

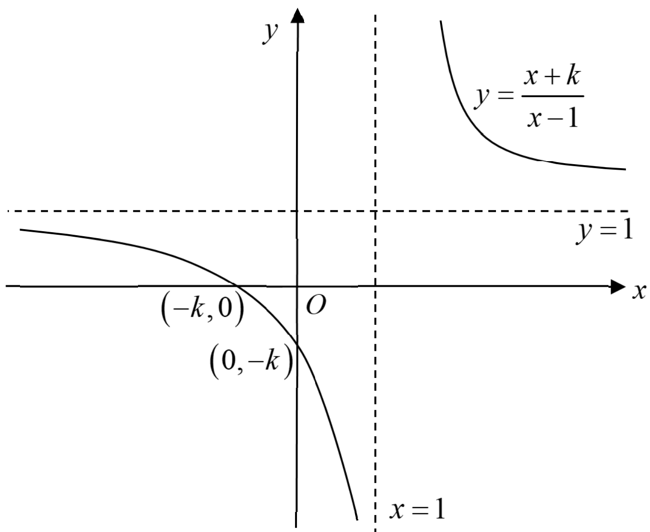
$x = \dfrac{y+k}{y-1}$

$\therefore \mathrm{g}^{-1}(x) = \dfrac{x+k}{x-1}$

$\mathrm{D_f} = \mathrm{R_f} = \mathrm{D_{f^{-1}}} = \mathbb{R} \setminus \{1\}$ ⟵ Should check that $\mathrm{D_f} = \mathrm{D_{f^{-1}}}$.

Since $\mathrm{g}(x) = \mathrm{g}^{-1}(x)$ for all *x* in the domain of g, g is self-inverse. (Shown)

(ii)



You need to first obtain the 2 asymptotes before sketching the curve.
Method:
By observation (for bilinear graphs) or long division/juggling

(iii)

Line of symmetry: $y = x$

Objective: $y = \dfrac{1}{x} \longrightarrow \ldots \longrightarrow y = \dfrac{x+k}{x-1} = 1 + \dfrac{k+1}{x-1}$

Perform long division for ease of transformation.

$y = \dfrac{1}{x}$

$\downarrow$ 1) Translation parallel to *x*-axis by 1 unit in the positive *x*-direction.

$y = \dfrac{1}{x-1}$

$\downarrow$ 2) Scale parallel to *y*-axis by factor $(k+1)$.

$\dfrac{y}{k+1} = \dfrac{1}{x-1}$

$y = \dfrac{k+1}{x-1}$

	$\downarrow$ 3) Translation parallel to y-axis by 1 unit in the positive y-direction. $y-1=\dfrac{k+1}{x-1}$ $y=1+\dfrac{k+1}{x-1}=\dfrac{x+k}{x-1}$

8	**Solutions**
(i)	$x-y=(x+y)^2$ $1-\dfrac{dy}{dx}=2(x+y)\left(1+\dfrac{dy}{dx}\right)$ $1-\dfrac{dy}{dx}=2(x+y)+2(x+y)\dfrac{dy}{dx}$ $(2x+2y+1)\dfrac{dy}{dx}=1-2x-2y$ $\dfrac{dy}{dx}=\dfrac{1-2x-2y}{2x+2y+1}$ $1+\dfrac{dy}{dx}=1+\dfrac{1-2x-2y}{2x+2y+1}$ $=\dfrac{1+2x+2y+1-2x-2y}{2x+2y+1}$ $=\dfrac{2}{2x+2y+1}$ (shown) You need **NOT** make y the subject before performing differentiation. Use implicit differentiation (especially if the result to be shown is a mixture of x and y variables).
(ii)	$\dfrac{d^2y}{dx^2}=(2)(-1)(2x+2y+1)^{-2}\left(2+2\dfrac{dy}{dx}\right)$ $=\left[\dfrac{-2}{(2x+2y+1)^2}\right](2)\left(1+\dfrac{dy}{dx}\right)$ $=-\left(\dfrac{2}{2x+2y+1}\right)^2\left(1+\dfrac{dy}{dx}\right)$ $=-\left(1+\dfrac{dy}{dx}\right)^2\left(1+\dfrac{dy}{dx}\right)$ $=-\left(1+\dfrac{dy}{dx}\right)^3$ (shown) Use implicit differentiation again.
(iii)	At turning point, $\dfrac{dy}{dx}=0$. $\dfrac{d^2y}{dx^2}=-(1+0)^3=-1$ Since $\dfrac{d^2y}{dx^2}<0$, the turning point is a maximum point. "Hence" question, so must use what was shown before in part **(ii)** to answer the question using second derivative test.

9	Solutions

(i)

$$\overrightarrow{AB}=\overrightarrow{OB}-\overrightarrow{OA}=\begin{pmatrix}-1\\-8\\1\end{pmatrix}-\begin{pmatrix}7\\8\\9\end{pmatrix}=\begin{pmatrix}-8\\-16\\-8\end{pmatrix}=-8\begin{pmatrix}1\\2\\1\end{pmatrix}$$

Vector equation of line AB: $\mathbf{r}=\begin{pmatrix}7\\8\\9\end{pmatrix}+\lambda\begin{pmatrix}1\\2\\1\end{pmatrix},\ \lambda\in\mathbb{R}$

> **Alternatively,** can write
> $\mathbf{r}=\begin{pmatrix}-1\\-8\\1\end{pmatrix}+\mu\begin{pmatrix}1\\2\\1\end{pmatrix},\ \mu\in\mathbb{R}$

(ii)

$\overrightarrow{ON}=\begin{pmatrix}7+\lambda\\8+2\lambda\\9+\lambda\end{pmatrix}$ for some $\lambda\in\mathbb{R}$

> *N* must be on the line *AB*

$$\overrightarrow{CN}=\begin{pmatrix}7+\lambda\\8+2\lambda\\9+\lambda\end{pmatrix}-\begin{pmatrix}1\\8\\3\end{pmatrix}=\begin{pmatrix}6+\lambda\\2\lambda\\6+\lambda\end{pmatrix}$$

$\overrightarrow{CN}\perp$ the line AB

$$\Rightarrow\begin{pmatrix}6+\lambda\\2\lambda\\6+\lambda\end{pmatrix}\cdot\begin{pmatrix}1\\2\\1\end{pmatrix}=0$$

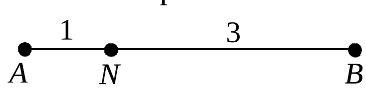
$\Rightarrow 6+\lambda+4\lambda+6+\lambda=0$

$\Rightarrow \lambda=-2$

Thus $\overrightarrow{ON}=\begin{pmatrix}5\\4\\7\end{pmatrix}$

$$\overrightarrow{AN}=\overrightarrow{ON}-\overrightarrow{OA}$$
$$=\begin{pmatrix}5\\4\\7\end{pmatrix}-\begin{pmatrix}7\\8\\9\end{pmatrix}=\begin{pmatrix}-2\\-4\\-2\end{pmatrix}=\frac{1}{4}\begin{pmatrix}-8\\-16\\-8\end{pmatrix}=\frac{1}{4}\overrightarrow{AB}$$

$\therefore AN:NB=1:3$

> Sketch a diagram if it helps to visualise the points on the line.
>
> 
>

(iii)

Let C' be the point of reflection of C in line AB.

$$\overrightarrow{AC'}=\overrightarrow{AC}+2\overrightarrow{CN}=\begin{pmatrix}1\\8\\3\end{pmatrix}-\begin{pmatrix}7\\8\\9\end{pmatrix}+2\begin{pmatrix}4\\-4\\4\end{pmatrix}=\begin{pmatrix}2\\-8\\2\end{pmatrix}=2\begin{pmatrix}1\\-4\\1\end{pmatrix}$$

Note that A lies on the line of reflection.

Cartesian equation of line of reflection is $x-7=\dfrac{y-8}{-4}=z-9$.

> You can use ratio theorem as well:
> $\overrightarrow{AN}=\dfrac{\overrightarrow{AC}+\overrightarrow{AC'}}{2}$
> $\Rightarrow\overrightarrow{AC'}=2\overrightarrow{AN}-\overrightarrow{AC}.$

10	Solutions
(i)	Let S be the external surface area of the model.

$S=\pi r^2+2\pi rh+\frac{1}{2}\left(4\pi r^2\right)$

Both h and r are variables. Thus, we should simplify the expression for S to either of r or h only. For ease of working, we will express S in terms of r only (which is useful for part **(ii)**).

Volume of model $=\pi r^2h+\frac{1}{2}\left(\frac{4}{3}\pi r^3\right)=k$

$h=\frac{1}{\pi r^2}\left(k-\frac{2}{3}\pi r^3\right)$

Sub $h=\frac{1}{\pi r^2}\left(k-\frac{2}{3}\pi r^3\right)$ into S:

$$S=\pi r^2+2\pi r\left[\frac{1}{\pi r^2}\left(k-\frac{2}{3}\pi r^3\right)\right]+2\pi r^2$$

$$=3\pi r^2+\frac{2}{r}\left(k-\frac{2}{3}\pi r^3\right)$$

$$=3\pi r^2+\frac{2k}{r}-\frac{4}{3}\pi r^2$$

Note:
k is a constant.

$$=\frac{5}{3}\pi r^2+\frac{2k}{r} \qquad (*)$$

For S to be minimum,

$$\frac{\mathrm{d}S}{\mathrm{d}r}=\frac{10}{3}\pi r-\frac{2k}{r^2}=0$$

$$\frac{10}{3}\pi r=\frac{2k}{r^2}$$

$$r^3=\frac{3k}{5\pi}$$

$$r=\left(\frac{3k}{5\pi}\right)^{\frac{1}{3}}$$

When $r=\left(\frac{3k}{5\pi}\right)^{\frac{1}{3}}$,

Always check that value is a minimum/maximum according to the question.

$$\frac{\mathrm{d}^2A}{\mathrm{d}r^2}=\frac{10}{3}\pi+\frac{4k}{r^3}$$

$$=\frac{10}{3}\pi+\frac{4k}{\left(\frac{3k}{5\pi}\right)}=10\pi>0 \quad \Rightarrow A \text{ is minimum}$$

	$h=\dfrac{1}{\pi\left(\dfrac{3k}{5\pi}\right)^{\frac{2}{3}}}\left(k-\dfrac{2}{3}\pi\left(\dfrac{3k}{5\pi}\right)\right)$ $=\dfrac{1}{\pi}\left(\dfrac{3k}{5\pi}\right)^{-\frac{2}{3}}\left(\dfrac{3k}{5}\right)$ $=\left(\dfrac{3k}{5\pi}\right)^{\frac{1}{3}}$
(ii)	Given $S=180$ and $V=k=200$, from (*) $180=\dfrac{5}{3}\pi r^2+\dfrac{2(200)}{r}$ $540r=5\pi r^3+1200$ $5\pi r^3-540r+1200=0$ Using GC, $r=-6.7587$ or $r=3.7215$ or $r=3.0372$ Leave intermediate working to 5 s.f. for greater degree of accuracy Since r denotes the radius, it must be positive. Hence, there are only two possible values of r: $r=3.7215$ or $r=3.0372$. When $r=3.7215$, $h=\dfrac{180-3\pi(3.7215)^2}{2\pi(3.7215)}=2.1157$ When $r=3.0372$, $h=\dfrac{180-3\pi(3.0372)^2}{2\pi(3.0372)}=4.8765$ Since $r<h$, $r=3.04, h=4.88$ Final answers to 3 s.f.

11	**Solutions**
(i)	$x=\theta-\sin\theta$ $\quad$ $y=1-\cos\theta$ $\dfrac{dx}{d\theta}=1-\cos\theta$ $\quad$ $\dfrac{dy}{d\theta}=\sin\theta$ $\dfrac{dy}{dx}=\dfrac{\sin\theta}{1-\cos\theta}$ $=\dfrac{2\sin\frac{1}{2}\theta\cos\frac{1}{2}\theta}{1-\left(1-2\sin^2\frac{1}{2}\theta\right)}$ Apply double angle formulae. (MF26) $=\dfrac{2\sin\frac{1}{2}\theta\cos\frac{1}{2}\theta}{2\sin^2\frac{1}{2}\theta}$ $=\dfrac{\cos\frac{1}{2}\theta}{\sin\frac{1}{2}\theta}$ $=\cot\frac{1}{2}\theta$ (shown)

<table>
<tr><td></td><td>

At $\theta = \pi$,

$\frac{dy}{dx} = \cot \frac{1}{2}\pi$

$= 0$

Gradient of C at the point where $\theta = \pi$ is 0.

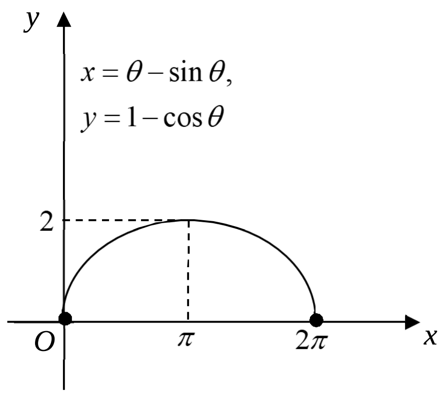
At $\theta = 0$, $\frac{dy}{dx}$ is undefined.

At $\theta = 2\pi$, $\frac{dy}{dx}$ is undefined.

The tangents become parallel to the y-axis as $\theta \to 0$ and $\theta \to 2\pi$.

Use the graph of $y = \cot x$ to visualise if need be.
Note: When gradient is undefined, the tangent is usually parallel to the y-axis.

</td></tr>
<tr><td>(ii)</td><td>



When sketching the curve, ensure that the gradients at $\theta = 0, \pi, 2\pi$ must be reflected according to part **(i)** correctly.

</td></tr>
<tr><td>(iii)</td><td>

when $x = 0, \theta = 0$

when $x = 2\pi, \theta = 2\pi$

Area $= \int_0^{2\pi} y \; dx$

$= \int_0^{2\pi} (1-\cos\theta)(1-\cos\theta) \; d\theta$

$= \int_0^{2\pi} (1-2\cos\theta+\cos^2\theta) \; d\theta$

$= \int_0^{2\pi} \left(1-2\cos\theta+\frac{1+\cos 2\theta}{2}\right) d\theta$

$= \left[\theta - 2\sin\theta + \frac{1}{2}\theta + \frac{1}{4}\sin 2\theta\right]_0^{2\pi}$

$= \frac{3}{2}(2\pi) - 2\sin 2\pi + \frac{1}{4}\sin 4\pi$

$= 3\pi$

Write down the integral with respect to x.

Do a substitution.

Apply double angle formulae. (MF26)

</td></tr>
<tr><td>(iv)</td><td>

Gradient of normal at P = $-\frac{1}{\cot\frac{1}{2}p} = -\tan\frac{1}{2}p$

Equation of normal at P is

$y-(1-\cos p) = -\tan\frac{1}{2}p\left[x-(p-\sin p)\right]$

</td></tr>
</table>

When $y = 0$,

$$\cos p - 1 = -\tan\frac{1}{2}p\left[x - \left(p - \sin p\right)\right]$$

$$x - p + \sin p = \frac{\left(1 - 2\sin^2\frac{1}{2}p\right) - 1}{-\tan\frac{1}{2}p}$$

Apply double angle formulae. (MF26)

$$x - p + \sin p = 2\sin\frac{1}{2}p\cos\frac{1}{2}p$$

$$x - p + \sin p = \sin p$$

$$x = p$$

Therefore, the normal to C at P crosses the x-axis at $(p, 0)$.

<u>Alternative solution</u>

Since normal to C at P crosses the x-axis, $y = 0$.

$$\cos p - 1 = -\tan\frac{1}{2}p\left[x - \left(p - \sin p\right)\right]$$

$$\cos p - 1 = -\frac{1 - \cos p}{\sin p}\left(x - p + \sin p\right)$$

$$\cos p - 1 = \frac{\cos p - 1}{\sin p}\left(x - p + \sin p\right)$$

$$\sin p = x - p + \sin p$$

$$x = p$$

Note:
From part **(i)**,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\sin\theta}{1 - \cos\theta} = \cdots = \cot\frac{1}{2}\theta$$

$$\therefore \frac{1 - \cos\theta}{\sin\theta} = \tan\frac{1}{2}\theta$$